CASE STUDY | DIGITAL TRANSFORMATION 5 MINUTE READ

Fonterra Co-operative Group Improves Processes

OEE calculations allow Fonterra employees to see data along with insights on any device, from anywhere, so they can make informed decisions.

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Fonterra Co-operative Group, a New Zealand-based multinational dairy co-operative, has a well-known portfolio of brands ranging from Anmum, a milk powder formula for mothers and infants, to Mammoth Supply Co., a lineup of individually bottled protein shakes and coffees.

Farm To Table

Owned by farmers in New Zealand, Fonterra employs thousands of employees around the world. Together, they collect around 22 billion liters of milk each season for their vast variety of products.

No matter the end-product, Fonterra's top priority across all brands is trust. Tankers transport raw milk from farms to 100 plants across 26 different manufacturing sites in New Zealand, where the raw milk is processed into bulk liquids.

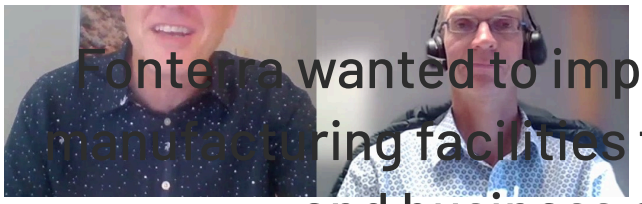
Ingredients and packaging materials for each unique product flow through various processing stages. Finished products are dispersed from distribution centers to customers or shipped to secondary manufacturing sites before finally being distributed to consumers across 140 countries.



Inspiring Trust

As the lead team for Fonterra's strategy, the Automation and Operational Technology (AOT) group needed an innovative and sustainable way to improve processes across manufacturing facilities, find an efficient way to securely deliver trustworthy data from plant floor to enterprise, implement an innovative, sustainable solution that revolutionized data collection and analytics capabilities, and inspire trust across manufacturing and enterprise functions.

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Fonterra wanted to improve processes across its manufacturing facilities to make critical operations and business decisions faster.

Fonterra needed to meet the growing demands of customers around the world, while ensuring its products were made with the same high quality in each facility. The AOT group knew that the breakthrough solution was in digital transformation and initiated a plan to find an efficient way to securely deliver trustworthy data from the plant floor to the enterprise.

Before embarking on this multiyear digital transformation journey, Fonterra's manufacturing data was a mixture of paper-based and electronic systems. At times, data could be incomplete or only understood by specific operators. Data could not be easily contextualized or shared with other facilities or the enterprise, and no centralized management process created version control issues.

Revolutionizing Data Collection

The team began the journey to implement an innovative, sustainable solution that could revolutionize data collection and analytics capabilities and inspire trust across manufacturing and enterprise functions.

Fonterra already used Rockwell Automation solutions to integrate closed loop analytics, manufacturing operations management (MOM), and track and trace solutions into its business. Now the AOT team would also leverage Rockwell Automation strategic alliances with Microsoft and PTC.

Before the transformation, Fonterra faced many challenges, including:

- Limited production capacity to meet customer demands, which meant the company might need to invest capital to build more plants
- A lack of standardization across its global plants
- Unplanned downtime due to events that were difficult to identify and troubleshoot to a speedy resolution
- Largely manual data collection and aggregation that lacked reliability, consistency, and context, which impacted the organization's ability to make informed, timely operational and business decisions

Fonterra's primary digital transformation goals included increasing OEE as part of continuous improvement across facilities and improving production capacity without the capital expenditure of building more plants. They also wanted to mitigate potential quality issues, reduce unplanned downtime, and democratize data to inform day-to-day operations, forecasting and strategic business decisions.

Overcoming Resistance

The initial phase took 3 years to complete and now empowers data-driven insights across more than 10 Fonterra sites. The team met some resistance to the new way of collecting and analyzing data. Accustomed to tracking day-to-day operations and yields using spreadsheets, some worried the automated digital system would misrepresent factors that required human experience to interpret.

The system proved itself, impressing stakeholders with the accuracy of the information and the ease with which they could now access, analyze, and share it. In practice, stakeholders were spending far less time crunching numbers and far more time using analytics to identify and inspire opportunities. As a result, the initiative reached a cultural turning point as plant managers zeroed in on areas for improvement and responded to errors faster, and executives make more agile decisions.

Rockwell Automation, Microsoft and PTC each brought their own best-in-class technologies and expertise and worked together to create a repeatable, scalable, and sustainable solution.

Solutions Combines Best of Three Companies

The digital transformation solution included Rockwell Automation control hardware and software, Microsoft cloud data sharing and storage products, and PTC application development platform designed for the Industrial Internet of Things (IIoT). This innovative edge-to-cloud-based solution connects information between plant operations and enterprise business teams at Fonterra through a single, trusted data environment.

The solution is built on the Rockwell Automation control platform using FactoryTalk® Metrics software, which provides OEE data to reveal ways to increase production,

reduce costs and increase quality. The OEE data is fed into PTC's ThingWorx application, part of FactoryTalk® InnovationSuite, which then delivers calculations to Microsoft's Power BI. This allows Fonterra employees at many levels within the business to visualize and share data across the organization and access it remotely.

The DX solution met and enabled all of Fonterra's primary goals to keep pace with market demands.

Fonterra now leverages Microsoft Power BI for daily production reports and system reviews to identify opportunities for optimization and to mitigate adverse operational events. In addition, Fonterra is in the planning stages of using Microsoft Azure to store data.

The digital transformation solution met and enabled all of Fonterra's primary goals. The average OEE score for Fonterra's plants has improved 20% and there's no longer a need to invest in building new facilities. While plant operators spend less time collecting and analyzing data, executives are more agile, making data-driven decisions to hold steady or pivot as necessary to keep pace with market demands.

Security Checkpoint

Before the solution could be fully implemented, Fonterra's security team conducted a rigorous security and network protocol review and approval process. Rigorous security measures had to be met, including to assign and manage access levels to internal stakeholders, and to thwart potential cybersecurity threats. The solution succeeded in securely connecting the automation floor to the enterprise.

Now, end-to-end from the plant floor through the enterprise, decision makers can access real-time data and analytics to monitor and assess operational effectiveness and business growth opportunities. They can more quickly diagnose and address unplanned events, such as equipment failures, and discern root causes to mitigate similar events, all while standardizing processes, quality and benchmarks across geographies.

Co-Innovation

The projects at Fonterra tested the larger co-innovation between Rockwell Automation, Microsoft, and PTC to create a digital transformation solution that could

be applied to other use cases in food and beverage, as well as other industries. To date, the companies have co-developed more than 20 use cases.

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